

Evaluating $\int e^{nt} \sin(mt) dt$ Real vs Complex Methods

Goal

We want to compute:

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- ▶ Integration by parts (real-variable method)
- ▶ Complex exponential method

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We compare efficiency and insight.

First Integration by Parts

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Choose:

$$u = \sin(mt), \quad dv = e^{nt} dt$$

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So:

$$I = \frac{1}{n} e^{nt} \sin(mt) - \frac{m}{n} \int e^{nt} \cos(mt) dt$$

Second Integration by Parts

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So:

$$J = \frac{1}{n} e^{nt} \cos(mt) + \frac{m}{n} I$$

Solve the System

Recall:

$$I = \frac{1}{n} e^{nt} \sin(mt) - \frac{m}{n} J$$

Substitute J :

$$I = \frac{1}{n} e^{nt} \sin(mt) - \frac{m}{n} \left(\frac{1}{n} e^{nt} \cos(mt) + \frac{m}{n} I \right)$$

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Simplify:

$$I + \frac{m^2}{n^2} I = \frac{1}{n} e^{nt} \sin(mt) - \frac{m}{n^2} e^{nt} \cos(mt)$$

Final Result (Real Method)

$$I = \frac{e^{nt}}{n^2 + m^2} (n \sin(mt) - m \cos(mt)) + C$$

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Observation:

- ▶ Requires two integrations
- ▶ Leads to a linear system
- ▶ Algebra is somewhat tedious

Key Idea

Use:

$$\sin(mt) = \Im(e^{imt})$$

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Then:

$$I = \Im \left(\int e^{nt} e^{imt} dt \right) = \Im \left(\int e^{(n+im)t} dt \right)$$

Single Integration

$$\int e^{(n+im)t} dt = \frac{1}{n+im} e^{(n+im)t}$$

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So:

$$I = \Im \left(\frac{e^{(n+im)t}}{n+im} \right)$$

Simplify the Expression

Multiply numerator and denominator:

$$\frac{1}{n + im} = \frac{n - im}{n^2 + m^2}$$

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Thus:

$$\frac{e^{(n+im)t}}{n + im} = \frac{e^{nt}}{n^2 + m^2} (n - im)(\cos mt + i \sin mt)$$

Take the Imaginary Part

Compute:

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Thus:

$$I = \frac{e^{nt}}{n^2 + m^2} (n \sin(mt) - m \cos(mt)) + C$$

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- ▶ **Integration by Parts:**
 - ▶ Mechanical
 - ▶ Requires iteration
 - ▶ Produces system of equations

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- ▶ **Integration by Parts:**
 - ▶ Mechanical
 - ▶ Requires iteration
 - ▶ Produces system of equations
- ▶ **Complex Method:**
 - ▶ Single computation
 - ▶ Uses exponential structure
 - ▶ Conceptually cleaner

Big Picture

- ▶ Complex exponentials unify sine and cosine
- ▶ Algebra replaces repeated integration
- ▶ Method generalizes easily

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Key takeaway:

Complex methods simplify real integrals